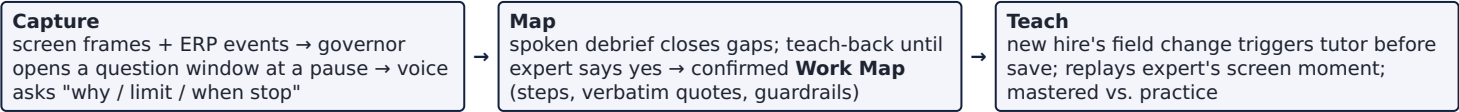


Simon — the AI Apprentice

Hack-Nation × ElevenLabs · **Capture** → **Map** → **Teach** · Live: tacit-ai-apprentice.vercel.app · Code: github.com/Backpacked333/sorcerer-apprentice-

Challenge Tackled

- **Problem:** a screen recording shows what an expert clicked, not why, when the rule stops applying, or who to ask. That judgment leaves with the person.
- **Users:** the expert (shares screen, works real cases) and the new hire (works an unseen case with the expert's reasoning as tutor). Demo domain: accounts-payable invoices in a sandbox ERP.
- **Approach:** an apprentice is a slot-filling machine, not a recorder. Each rule has empty slots (reason, limit, exception, who to ask). The screen shows when a judgment happened; voice fills a slot only at a pause. "Understood" = zero open slots **plus** the expert's explicit yes.



Tools / ML Models Used

ElevenAgents (V4 Turbo voice, Gemini 2.5 Flash)	Two voice agents: interviewer and tutor. Mic is closed outside governor-opened windows; speaks only on tagged messages, via client tools (log_answer, record_prediction).
ElevenLabs Scribe v2 Realtime	Live speech-to-text of the expert; transcript is the only source of verbatim quotes.
Claude Haiku 4.5 (vision)	Turns screen stills into events labeled vision; sandbox ERP telemetry is labeled dom — two sources, never disguised as one.
Claude Sonnet 4.5 (compile)	Compiles transcript + events into the Work Map (structured output); a deterministic compile is the keyless fallback. A model never confirms its own map.
Next.js 16, React 19, Tailwind 4, Zod 4, AI SDK 7 / Vercel AI Gateway, Supabase (Postgres + private Storage), Vercel, Vitest	App, schemas, model routing, per-visitor isolated storage, hosting, tests.

What Worked Well

- **Timing is code, not a prompt.** A deterministic governor decides when the agent may speak; in production tests, questions began in ~585-740 ms on one connection and closed gates stayed muted.
- **Honesty by construction:** verbatim-or-nothing quotes, rules only where the expert stated the trigger, labeled fallbacks, consent-epoch "off the record" that strikes transcript, events, frames and clips.
- **Only a confirmed map teaches:** tutor, save guard and exports load only maps with an expert confirmation.
- **Keyless mode** runs the full loop (seed → Capture → Map → confirm → tutor → guard) as an end-to-end smoke test; 546 automated tests, typecheck and production build pass.

What Was Challenging

- **Voice reliability:** background voices and agent echo triggered false answers. Fix: background-voice detection, a required log_answer tool confirmation, and discarding pre-listening transcript (false-quote bug P-12).
- **SDK drift:** the installed ElevenLabs SDK dropped a VAD field and had a stale model enum; we override the wire payload and verify via read-back (agents:create --check).
- **Self-disconnect on timeout** was mistaken for user hang-up; now an 8 s wait before a labeled browser-speech fallback.
- **Trust:** masks painted before frames leave the browser; strict evidence-clock checks so one invoice's narration cannot satisfy another's.
- **Limits:** human listening/real-room naturalness acceptance is still pending; we did not claim it.

How We Spent Our Time (approx., from git history; 4 people, parallel lanes)

Sat 1 PM-5 PM	Brief → spec, contracts, lanes (Voice · Eyes/Trust · Map/Teach · Experience), keyless baseline
Sat 5 PM-11 PM	Voice adapter and governor, vision + redaction, compile and teach-back, ERP sandbox; first agent provisioning
Sat 11 PM-Sun 3 AM	Integration: CaptureLoop, Supabase + Vercel deploy, V4 Turbo release, voice-quality and evidence-integrity fixes
Sun 3 AM-7:30 AM	Feature freeze, production tests, three 60 s videos, team photo, this report, submission

If we had 24 more hours, we'd run real human listening sessions with strangers as the expert and the new hire, tune the pause governor on those recordings, and extend the Work Map beyond accounts-payable to a second domain.